

Level and question	Unit read	Statistic and pass rule	Comparator	A pass allows
LEVEL 4 Structural fidelity Does the scale hang together the way it does in people, and the same way across groups?	All draws of one model and framing, as one population; personas resampled	One-factor model on polychoric correlations; congruence with the NHANES loadings; invariance across sex, race and income. Pass: in both framings, a general factor (every loading $\geq .30$, first-to-second eigenvalue ratio ≥ 3), congruence $\geq .95$ at its lower 90% limit and loading RMSD $\leq .10$ at its upper limit, and every invariance step agreeing with NHANES at margin .08. The factor within personas is reported, not scored.	NHANES 2005–2018 loadings and invariance verdicts	Treat the total as one score that means the same in every group
LEVEL 3 Population calibration Is each group's level right?	Persona means of k draws, reweighted to each NHANES group's makeup	Simulated minus NHANES mean PHQ-8, for all adults and for nine groups by race, sex and income. Pass: every 90% interval within ± 2 points (0.2 SD and 1 point also reported). The share scoring 10 or more is a second row.	NHANES 2005–2018 group means (2021–2023 beside them)	Use the simulated level as a stand-in for the population's
LEVEL 2 Subgroup fidelity Are the gaps between groups the right size and direction?	Persona means of k draws, differenced within pairs that differ on one attribute	Ratio of the simulated gap to the NHANES gap, with a 98.6% interval (Bonferroni over seven contrasts). Pass: the interval lies within 0.75 to 1.25. Each contrast is kept, not kept (interval outside 0.75 to 1.25) or unresolved (interval overlapping it). Not read: no population gap; reference too imprecise to certify kept.	NHANES 2005–2018 gap with the other persona attributes held equal (crude gap beside it)	Read the size and direction of group differences
LEVEL 1 Individual coherence Does a single draw answer like one person?	Single draw: one answer vector, with inference clustered by persona	Person fit of the item pattern under a graded response model fitted to NHANES, compared with adults at the same total. Pass: the shares of draws below the NHANES 5th and above its 95th percentile, each divided by 5%, have 90% intervals within 0.67 to 1.5.	NHANES 2005–2018 adults with the same PHQ-8 total	Use a single draw as one case, such as an exemplar or a test item <i>A failure rules out single-draw uses only. Levels 2 to 4 still run.</i>
GATE Regeneration stability Are persona scores stable enough to read?	Repeated draws of one persona, within one model and framing	Variance of scores between personas and between draws; dependability of a mean of k draws. Pass: k is the fewest draws giving $SE \leq 0.125$ reference SD in each framing (minimum rule 0.25 SD). A single draw passes with $SD \leq 0.25$ reference SD. Dependability is reported as persona separation.	None: draws are compared with each other	Read persona means of k draws at levels 2 and 3

The gate fixes k , the number of draws averaged into each persona mean, for levels 2 and 3. Apart from this, every level runs whatever the result below it.